



International Islamic University Chittagong

Department of Computer Science and Engineering

Course Name: Computer Graphics Lab

Course Code: CSE-4742

Submitted To:

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The background features several abstract blue watercolor-like shapes. In the top left, there are two overlapping, soft-edged shapes. In the bottom left, there is a cluster of small, dark blue dots. On the right side, there are several thin, curved, parallel lines that sweep upwards, and a larger, soft-edged shape at the bottom right.

Multiple 3D Games

Here is where my presentation begins

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To develop a **multi-game platform** using OpenGL

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To implement **3D graphics and real-time rendering**

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To design **interactive gameplay** using keyboard input

04

To integrate **sound and visual effects**

05

To apply **computer graphics concepts** in game development

The background of the slide features abstract blue watercolor splashes and line art. On the left, there are several curved, parallel blue lines. In the center and right, there are irregular, soft-edged blue shapes of varying sizes, some with darker blue dots scattered around them, resembling a stylized galaxy or nebula.

Introduction

- This project is a **3D graphics-based gaming application**.
- It combines multiple classic games into **one single platform**.
- Developed using **C++ with OpenGL and GLUT**.
- Focuses on **graphics rendering, animation, collision detection, and interaction**.
- Provides an engaging and educational gaming experience.

The background of the slide features abstract blue watercolor splashes and line art. On the left, there are several curved, parallel lines. In the center and right, there are irregular, organic shapes in various shades of blue, some with smaller dots scattered around them.

Objectives

- The main objectives of this project are:
- To develop a multi-game platform using OpenGL and GLUT
- To implement 3D graphics rendering with lighting and depth
- To design interactive gameplay with keyboard
- To integrate sound effects for better user experience
- To apply particle systems for explosion and visual effects



Tools and Technologies

- **Programming Language:**
C++
- **Graphics_Library:** OpenGL
- **Utility Toolkit:** GLUT
- **Development_Environment:**
Code::Blocks
- **Platform:** Windows

Main Features

- 🎮 Multiple games in one application
- 📦 3D graphics with lighting and depth
- 🔊 Sound effects for game actions
- 💥 Particle effects for shooter game
- ⌨️ Keyboard-based controls
- 🧠 Collision detection & score system
- 💻 Menu-driven user interface

Methodology

Methodology

The project follows the steps below:

Requirement Analysis

- Select classic games suitable for 3D conversion
- Identify required controls & game logic



Graphics Setup

- Initialize OpenGL context using GLUT
- Configure lighting, camera, and projection



Game Design

- Create separate logic for Snake, Pong, Shooter, and Tetris
- Implement collision detection and scoring system



Sound Integration

- Add sound effects using system commands



Rendering & Animation

- Use real-time rendering loop
- Apply transformations, lighting and particle effects

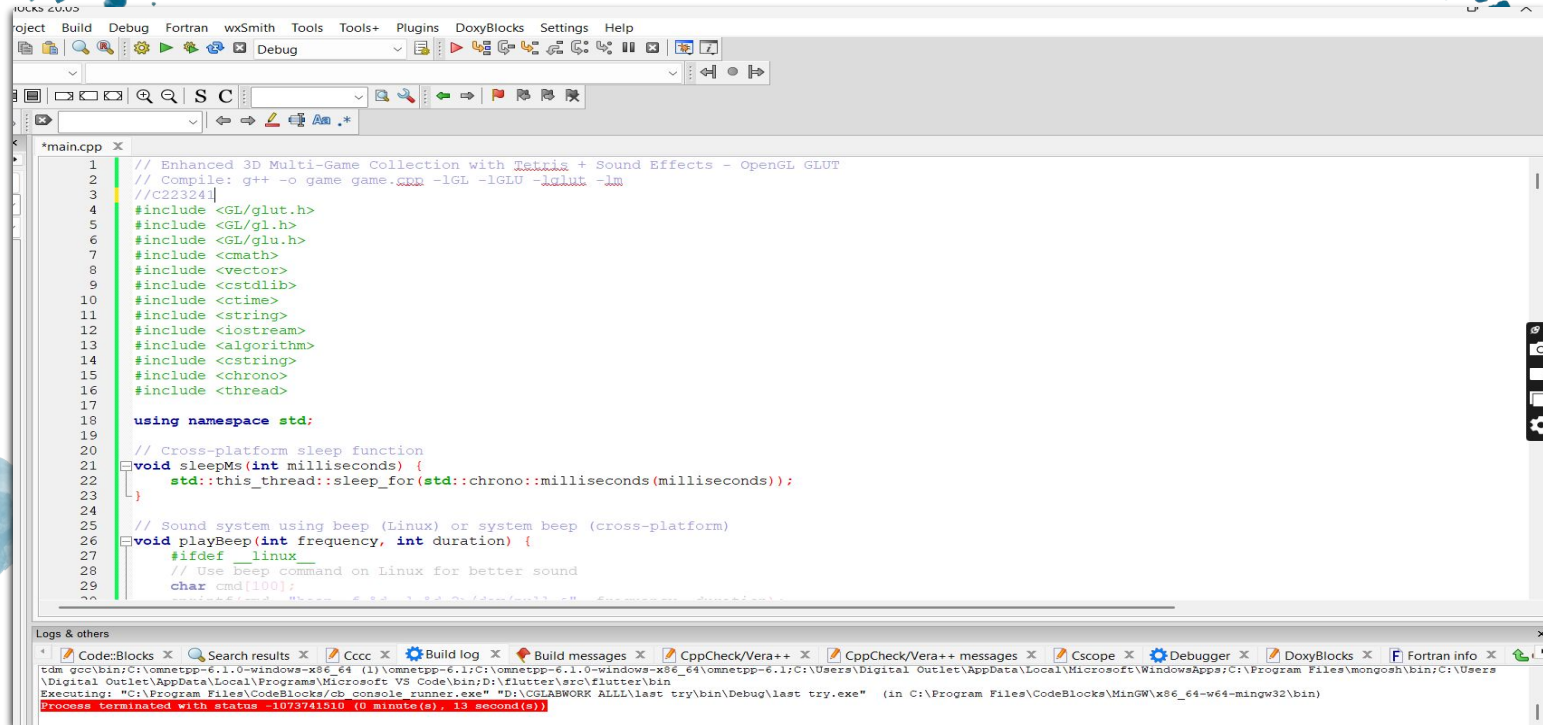


Testing & Optimization

- Test gameplay smoothness
- Fix collision and performance issues



Code & Necessary Screenshots



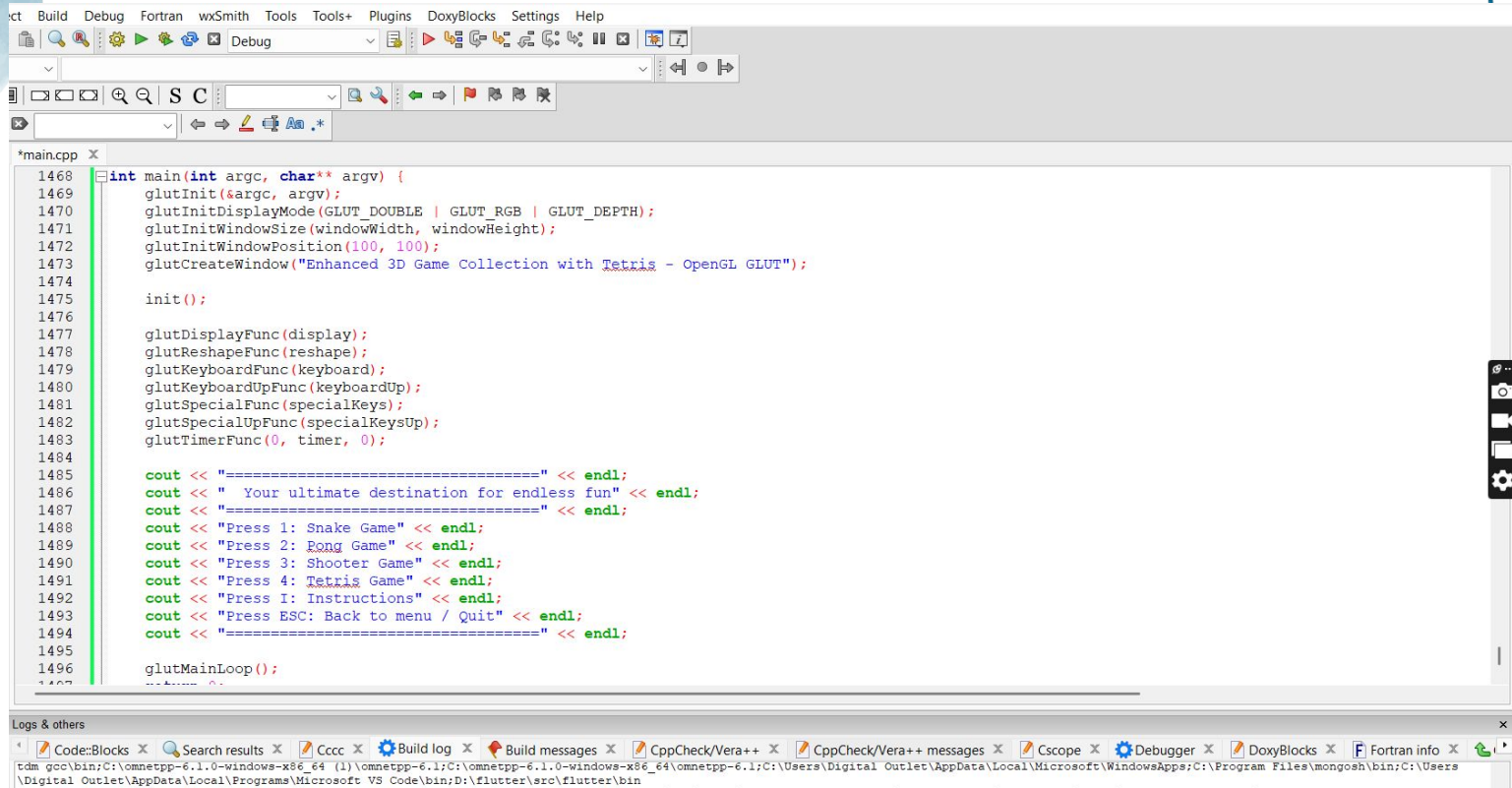
The screenshot displays a C++ code editor with a menu bar (Project, Build, Debug, Fortran, wxSmith, Tools, Tools+, Plugins, DoxyBlocks, Settings, Help) and a toolbar. The main window shows the file `*main.cpp` with the following code:

```
1 // Enhanced 3D Multi-Game Collection with Tetris + Sound Effects - OpenGL GLUT
2 // Compile: g++ -o game game.cpp -lGL -lGLU -lglut -lm
3 //C223241
4 #include <GL/glut.h>
5 #include <GL/gl.h>
6 #include <GL/glu.h>
7 #include <cmath>
8 #include <vector>
9 #include <cstdlib>
10 #include <ctime>
11 #include <string>
12 #include <iostream>
13 #include <algorithm>
14 #include <cstring>
15 #include <chrono>
16 #include <thread>
17
18 using namespace std;
19
20 // Cross-platform sleep function
21 void sleepMs(int milliseconds) {
22     std::this_thread::sleep_for(std::chrono::milliseconds(milliseconds));
23 }
24
25 // Sound system using beep (Linux) or system beep (cross-platform)
26 void playBeep(int frequency, int duration) {
27     #ifdef __linux__
28         // Use beep command on Linux for better sound
29         char cmd[100];
30         sprintf(cmd, "beep -f %d -d %d", frequency, duration);
31         system(cmd);
32     }
33 }
```

The bottom panel shows the 'Logs & others' window with the following output:

```
tdm gcc/bin/C:/omnetpp-6.1.0-windows-x86_64 (1) omnetpp-6.1.0/C:/omnetpp-6.1.0-windows-x86_64/omnetpp-6.1.0/C:/Users/Digital Outlet/AppData/Local/Microsoft/WindowsApps/C:/Program Files/mongosh/bin/C:/Users/Digital Outlet/AppData/Local/Programs/Microsoft VS Code/bin/D:/flutter/bin/flutter/bin
Executing: "C:/Program Files/CodeBlocks/cb_console_runner.exe" "D:/CGIABWORK ALLI/last try/bin/Debug/last try.exe" (in C:/Program Files/CodeBlocks/MinGW/x86_64-w64-mingw32/bin)
Process terminated with status -1073741510 (0 minute(s), 13 second(s))
```

Fig:1. C++ Code.



```
1468 int main(int argc, char** argv) {
1469     glutInit(&argc, argv);
1470     glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH);
1471     glutInitWindowSize(windowWidth, windowHeight);
1472     glutInitWindowPosition(100, 100);
1473     glutCreateWindow("Enhanced 3D Game Collection with Tetris - OpenGL GLUT");
1474
1475     init();
1476
1477     glutDisplayFunc(display);
1478     glutReshapeFunc(reshape);
1479     glutKeyboardFunc(keyboard);
1480     glutKeyboardUpFunc(keyboardUp);
1481     glutSpecialFunc(specialKeys);
1482     glutSpecialUpFunc(specialKeysUp);
1483     glutTimerFunc(0, timer, 0);
1484
1485     cout << "===== " << endl;
1486     cout << "  Your ultimate destination for endless fun" << endl;
1487     cout << "===== " << endl;
1488     cout << "Press 1: Snake Game" << endl;
1489     cout << "Press 2: Pong Game" << endl;
1490     cout << "Press 3: Shooter Game" << endl;
1491     cout << "Press 4: Tetris Game" << endl;
1492     cout << "Press I: Instructions" << endl;
1493     cout << "Press ESC: Back to menu / Quit" << endl;
1494     cout << "===== " << endl;
1495
1496     glutMainLoop();
1497 }
```

Logs and others

* Code::Blocks X Search results X Cccc X Build log X Build messages X CppCheck/Vera++ X CppCheck/Vera++ messages X Cscope X Debugger X DoxyBlocks X Fortran info X

tdm gcc/bin;C:\omnetpp-6.1.0-windows-x86_64 (1)\omnetpp-6.1.0-windows-x86_64\omnetpp-6.1.0\Users\Digital Outlet\AppData\Local\Microsoft\WindowsApps\C:\Program Files\mongosh\bin;C:\Users\Digital Outlet\AppData\Local\Programs\Microsoft VS Code\bin;D:\flutter\src\flutter\bin

Fig.3. C++ Code.(contd.).

Result



Vedio of Game Startup Interface.

Snake Game

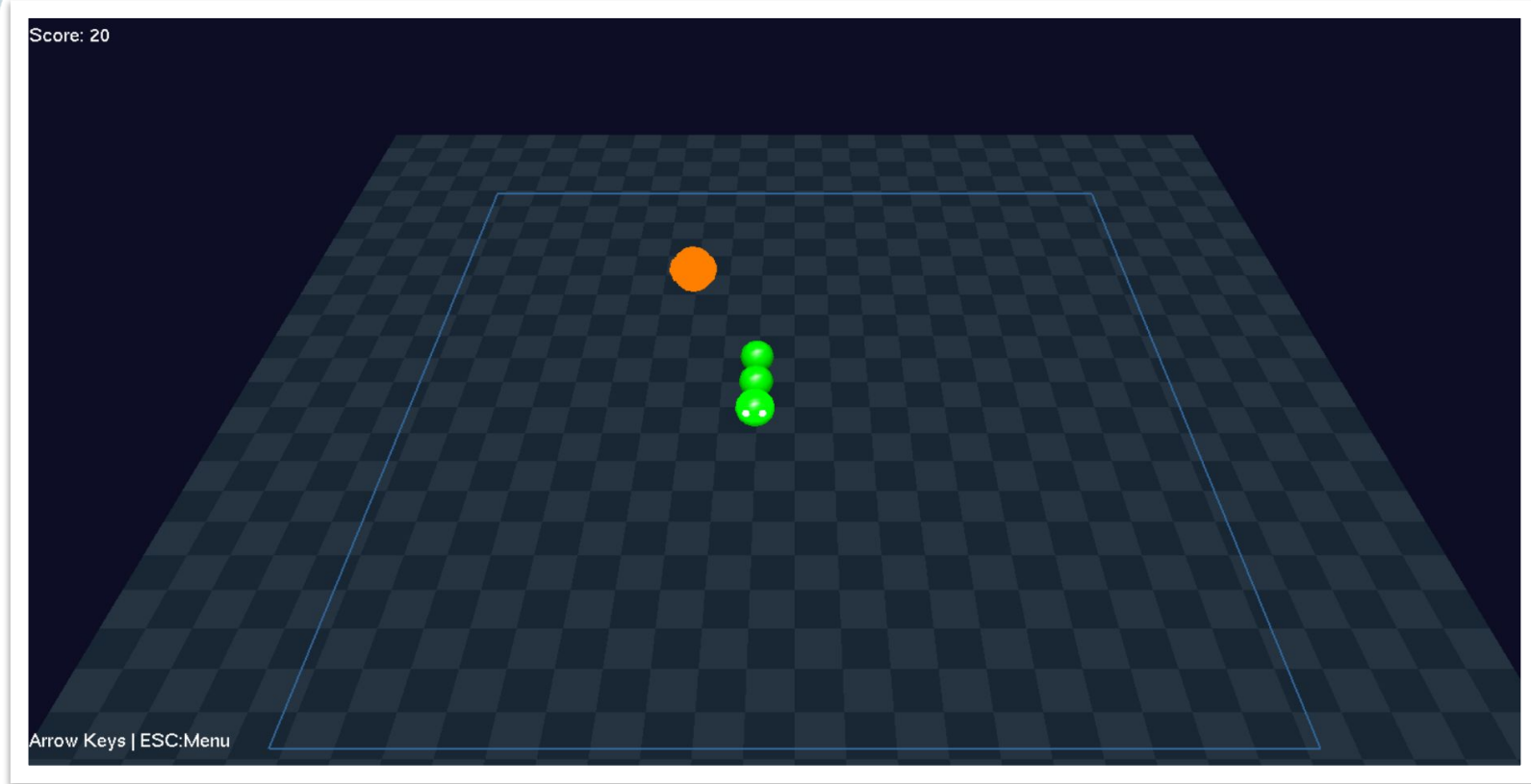


Fig:4. 3D Snake Game Interface.

Pong Game

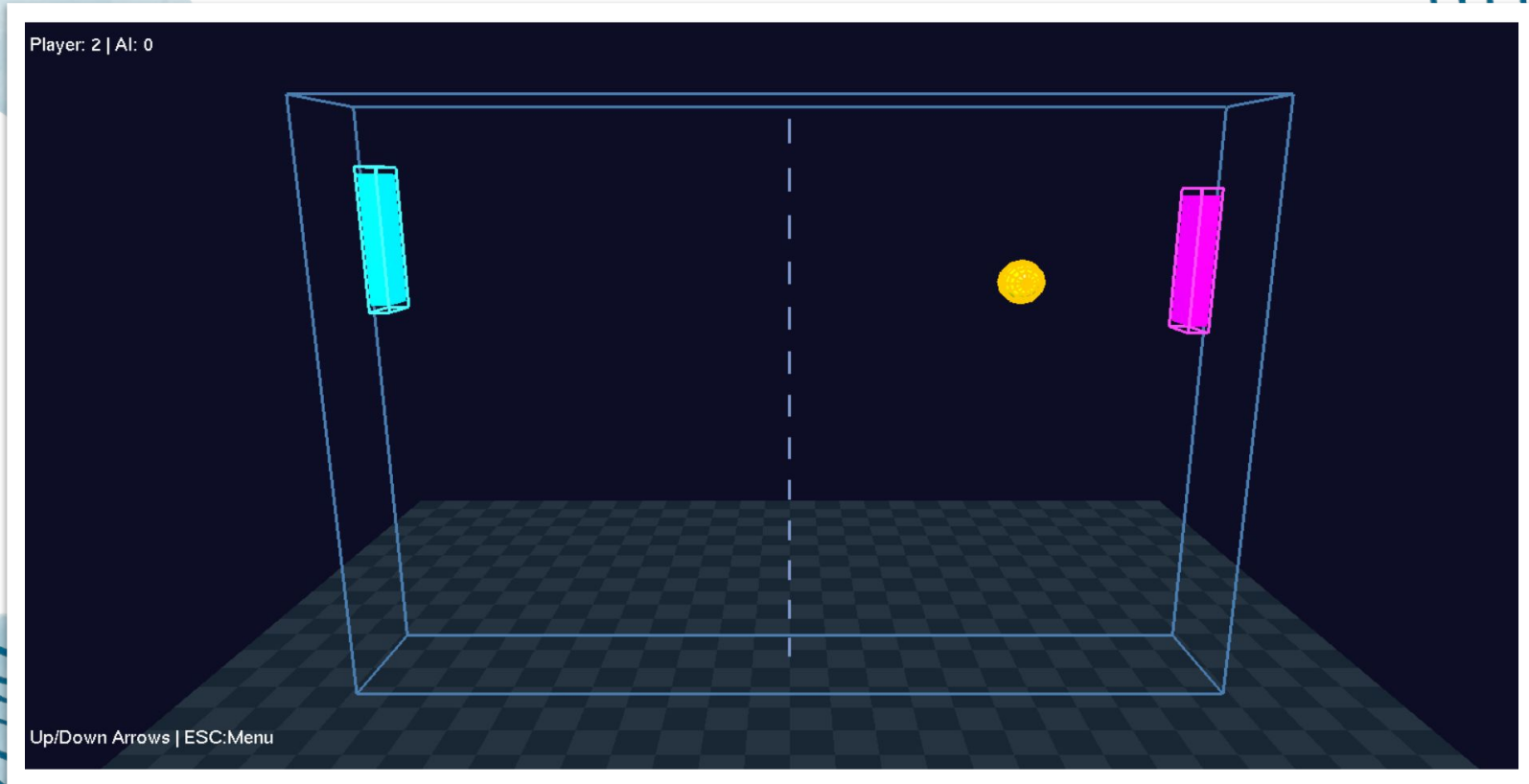


Fig:5. 3D Pong Game Interface.

Shooter Game

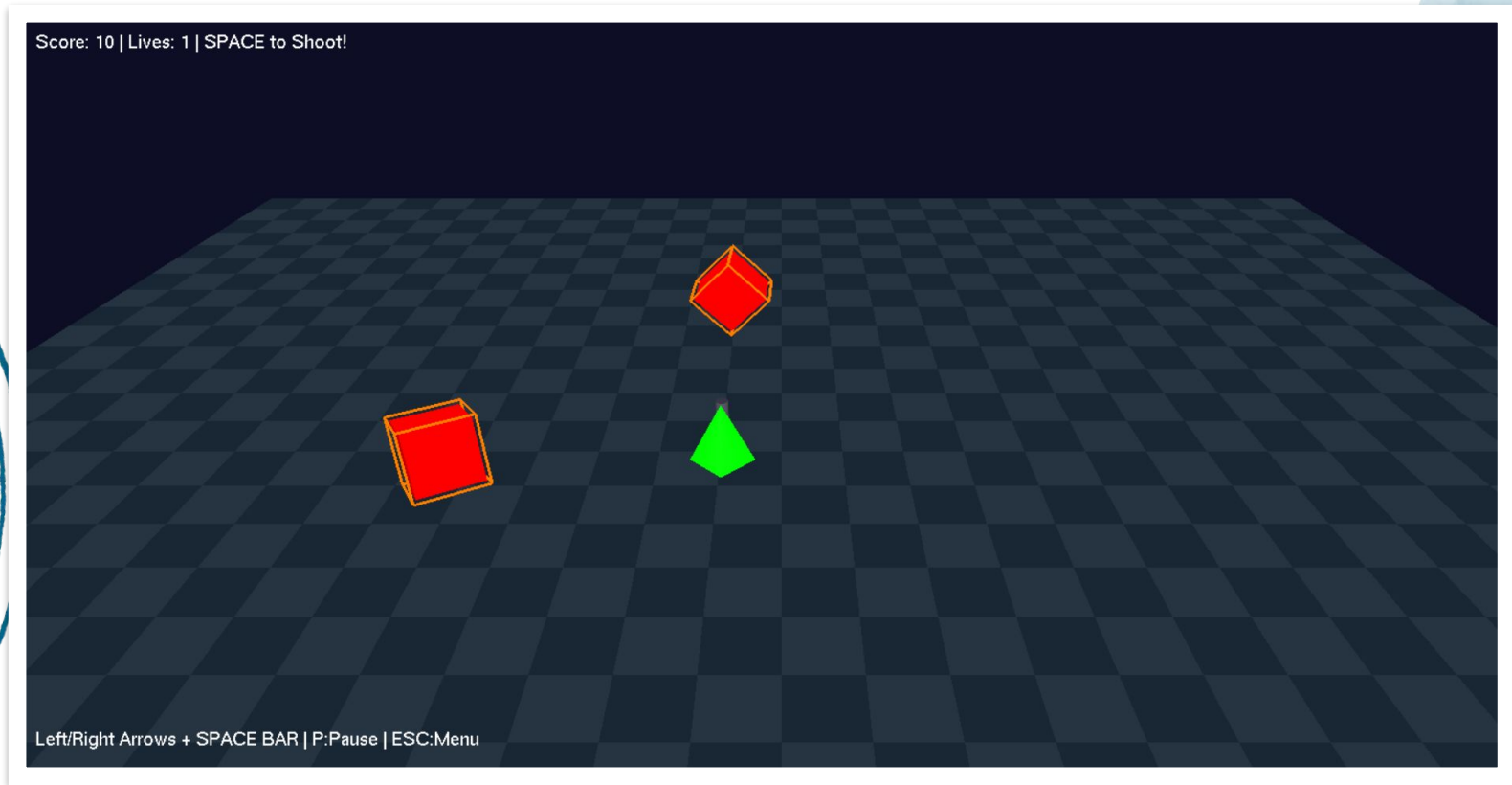


Fig:6. 3D Snake Game Interface

Tetris Game

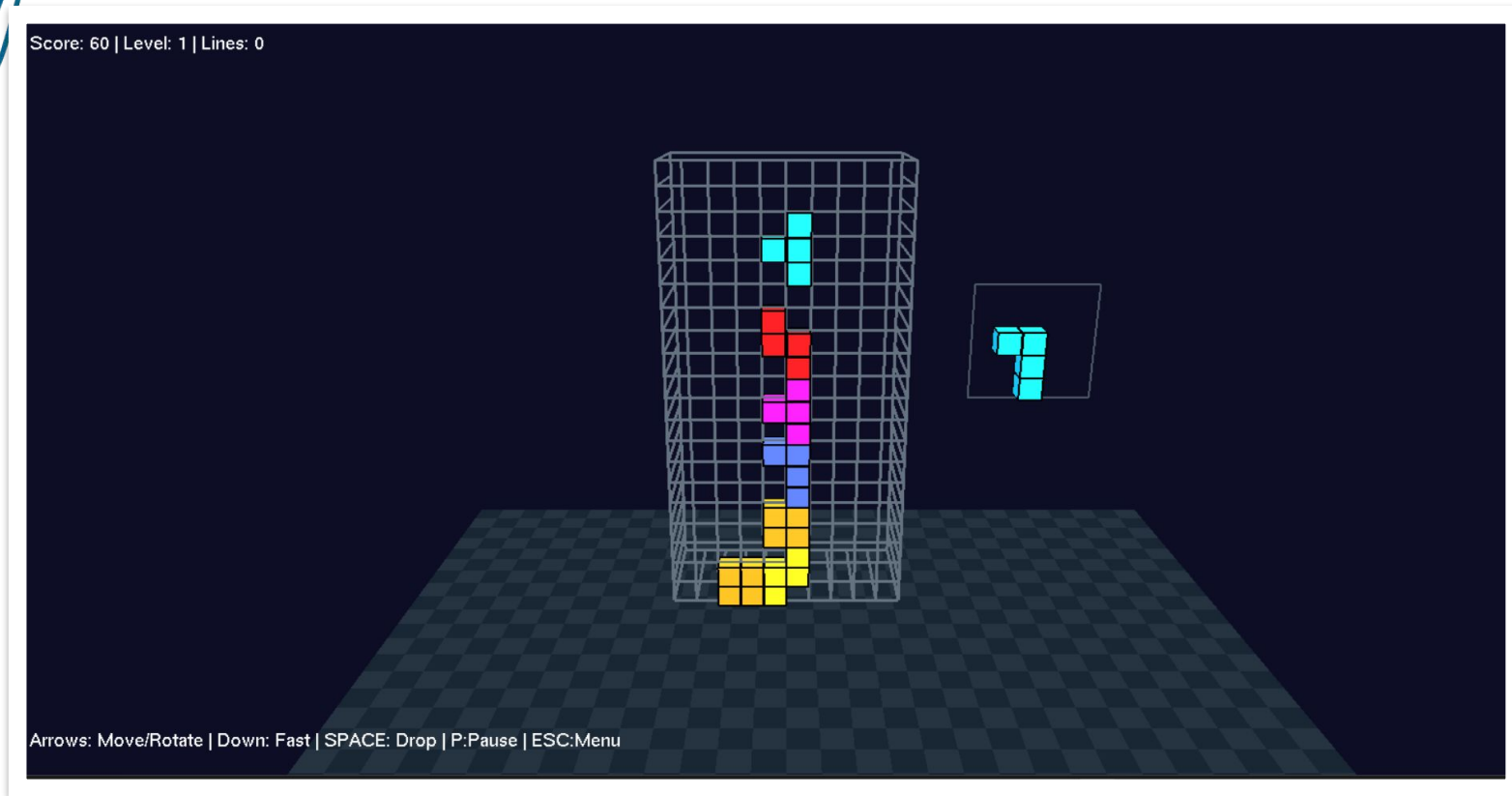


Fig:7 3D Tetris Game Interface.



Conclusion

The project of multiple 3D Game Collection project successfully demonstrates the use of OpenGL and GLUT in game development. The Snake and Tetris games were implemented with proper game logic and graphics rendering. This project improves understanding of computer graphics, real-time interaction, and game design concepts, making it suitable as an academic and practical project.



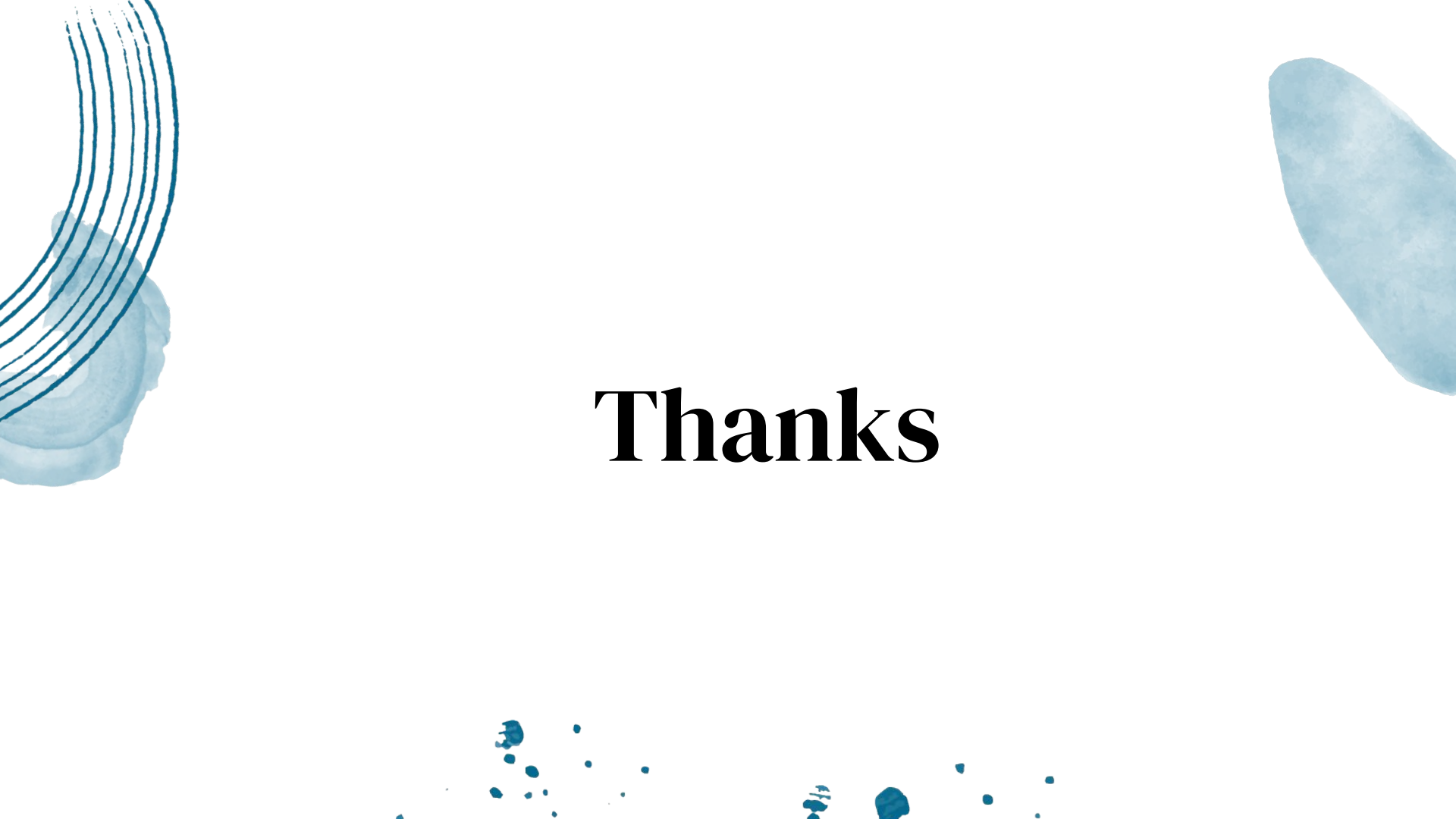
References

 **SnakeGameDemo**

<https://youtube.com/shorts/r5vzB8M-Q28?si=w0tg4Bq3k8YDC3gP>

 **TetrisGameDemo**

<https://youtube.com/shorts/-ZgvgwbbQiM?si=k3TVvzLJDWuEWN7p>



Thanks